

Complete Step-by-Step Guide to Install V2Ray/Xray

Prerequisites

1. What You Need:

Item	Description	Free/Affordable Options
Foreign Server	VPS outside restricted regions	DigitalOcean, Hetzner, Vultr, Oracle Cloud (always free)
Domain	Domain name for SSL	Freenom (free) or purchase from Namecheap
Cloudflare	CDN for obfuscation	Free account at cloudflare.com
Terminal	To connect to server	Windows: PuTTY, Mac/Linux: Terminal
Client App	Software for mobile/computer	v2rayNG (Android), v2rayN (Windows)

Step 1: Purchase and Set Up Server

A) Choose Suitable Server:

1. Go to sites like DigitalOcean.com
2. Create an account (may need international credit card)
3. Create a Droplet (virtual server):
 - o **Location:** Germany (Frankfurt) or Netherlands (Amsterdam)
 - o **OS:** Ubuntu 22.04 LTS
 - o **Plan:** Minimum 1GB RAM (~\$5 monthly)

B) Connect to Server:

After purchasing, you'll receive server IP and password.

On Windows (with PuTTY):

1. Download and install PuTTY
2. Enter server IP
3. Username: root
4. Password: Received password

On Mac/Linux (with Terminal):

```
bash
ssh root@IP-ADDRESS
# Then enter password
```

✂ Step 2: Install Xray (Better than V2Ray)

Completely Step-by-Step:

Step 1: Connect to server (as mentioned above)

Step 2: Copy this command and paste in server terminal (Right-click → Paste):

```
bash
apt update && apt upgrade -y
```

Step 3: Now run this command:

```
bash
bash -c "$(curl -L https://github.com/XTLS/Xray-install/raw/main/install-release.sh)"
@ install
```

Wait for installation to complete ✓

Step 4: Check if installed:

```
bash
xray version
```

If it shows version, installation was successful.

Step 3: Create Config File

Easy Method (Auto Generation):

Run this command:

```
bash
cd /usr/local/etc/xray
nano config.json
```

Now **copy** all content below and **paste in nano** (in nano: Right-click → Paste):

```
json
{
  "log": {
    "loglevel": "warning"
  },
  "inbounds": [
    {
      "port": 443,
      "protocol": "vless",
      "settings": {
        "clients": [
          {
            "id": "12345678-1234-1234-1234-123456789abc",
            "flow": "xtls-rprx-vision"
          }
        ]
      },
      "decryption": "none"
    },
    "streamSettings": {
      "network": "tcp",
      "security": "reality",
      "realitySettings": {
        "dest": "www.google.com:443",
        "serverNames": [
          "www.google.com",
```

```

        "www.cloudflare.com",
        "www.microsoft.com"
    ],
    "privateKey": "aK9Dq3jXpL8mFgR2sT5yV7wZbNcQ8xJ0z",
    "shortIds": [
        "6ba85179",
        "e30d4fc2"
    ]
    }
}
},
"outbounds": [
    {
        "protocol": "freedom",
        "tag": "direct"
    }
]
}

```

To save and exit:

1. Press `Ctrl + X`
2. Press `Y` (to confirm save)
3. Press `Enter`

🔑 Step 4: Generate Your Own Information

A) Create New UUID:

```

bash
cat /proc/sys/kernel/random/uuid

```

A code like this will be generated: 12345678-1234-1234-1234-123456789abc

Copy this code.

B) Create New privateKey:

```
bash
xray x25519
```

Output will be:

```
text
Private key: aK9Dq3jXpL8mFgR2sT5yV7wZbNcQ8xJ0z
Public key: bM8Er7kYq9P6nT4wV2xZ5cR8dF1gH3jL0p
```

Copy the Private key: line.

C) Edit File with Your Information:

Open the file again:

```
bash
nano /usr/local/etc/xray/config.json
```

Change these two places:

1. "id": "YOUR-UUID-HERE"
2. "privateKey": "YOUR-PRIVATE-KEY-HERE"

Then again `Ctrl+X` → `Y` → `Enter`

► Step 5: Start the Service

A) Start Xray:

```
bash
systemctl enable xray
systemctl start xray
```

B) Check Status:

```
bash
```

```
systemctl status xray
```

If it shows `active (running)`, it's OK ✓

C) If Error Occurs:

```
bash
```

```
journalctl -u xray -f
```

To see errors (exit with `Ctrl+C`)

Step 6: Install on Mobile (Android)

Install v2rayNG:

1. Go to github.com/2dust/v2rayNG
2. Download latest APK
3. Install on mobile (may need to enable `Install from unknown sources`)

Configure v2rayNG:

1. Open the app
2. Tap on **+**
3. Select **Manual Input**

Required Information from Server:

Fill with this information:

Field	Value
Address	Your server IP (e.g., 123.123.123.123)

Field	Value
Port	443
ID	The UUID you created
Flow	xtls-rprx-vision
Network	tcp
Security	reality
Public key	The Public key from x25519 (second line)
Short ID	6ba85179
SpiderX	Leave empty

Example:

```

text
Address: 123.123.123.123
Port: 443
ID: 12345678-1234-1234-1234-123456789abc
Flow: xtls-rprx-vision
Network: tcp
Security: reality
Public key: bM8Er7kYq9P6nT4wV2xZ5cR8dF1gH3jL0p
Short ID: 6ba85179

```

4. Tap **Save**
5. Tap the switch (turn blue)
6. Test internet connection

Step 7: Install on Windows

Install v2rayN:

1. Go to github.com/2dust/v2rayN
2. Download v2rayN.zip
3. Extract the file
4. Run v2rayN.exe

Configure v2rayN:

1. Right-click on v2rayN icon in system tray (near clock)
2. **Subscribe** → **Add Subscription**
3. In Address field, put this link:

text

```
vmess://eyJhZGQiOiIxMjMuMTIzLjEyMy4xMjMiLCJhawQiOiIwIiwiaWxwbiI6IiIsImZwIjoiIiwiaG9zdCI6IiIsImlkIjoiMTIzNDU2NzgtMTIzNC0xMjM0LTEyMzQtMTIzNDU2Nzg5YWJjIiwibmV0IjoidGNwIiwicGF0aCI6IiIsInBvcnQiOiI0NDMiLCJwcyI6IiIsInB1cnZlciI6IiIsInNjeSI6IiIsInJlYWxpdiI6IiIsInZiI6IiIsInR5b29nbGUuY29tIiwic25pIjoiNmJhODUxNzkiLCJ0bHMiOiIiLCJ0eXB1IjoiIm9uZSI6IiIsInYiOiIyIn0=
```

Note: In this link, replace with your own IP and UUID.

Simpler Method:

1. Click **Servers** → **Add VMESS Server**
2. Fill information same as mobile
3. Check **Allow Insecure** box
4. Click OK and double-click on the server

□ Step 8: Test Connection

On Server:

```
bash
```

```
# Is port 443 open?
```

```
netstat -tlnp | grep 443
```



```
# Is firewall open?
```

```
ufw status
```

```
# If firewall is active:
```

```
ufw allow 443/tcp
```

On Mobile/Computer:

1. Go to ipleak.net
 2. Should show IP of another country (not your country)
 3. Go to speedtest.net to test speed
-

Step 9: Troubleshooting

If Not Connecting:

1. Check Server:

```
bash
```

```
# Is Xray running?
```

```
systemctl status xray
```

```
# If not, restart:
```

```
systemctl restart xray
```

```
# See Logs:
```

```
journalctl -u xray -n 50
```

2. Check Port:

```
bash
```

```
# Test from outside (from your computer):
```

```
telnet IP-SERVER 443
```

```
# or
```

```
curl -v https://IP-SERVER
```

3. Common Problems:

Problem	Solution
Timeout	Port 443 might be blocked, try port 8443 or 2053
Certificate error	Check "Allow Insecure" in client
Protocol error	Check Flow should be <code>xtls-rprx-vision</code>
Slow speed	Choose different server or change from Reality to WebSocket

Step 10: More Professional Method (WebSocket + TLS)

If above method doesn't work, try this:

On Server:

```
bash
# First you need domain (e.g., myserver.com)
# Install SSL certificate:
apt install certbot -y
certbot certonly --standalone -d myserver.com
```

```
# New config file:
nano /usr/local/etc/xray/config-ws.json
```

Paste this content:

```
json
{
  "inbounds": [{
    "port": 443,
    "protocol": "vless",
    "settings": {
```

```
    "clients": [{
      "id": "YOUR-UUID",
      "flow": ""
    }]
  },
  "streamSettings": {
    "network": "ws",
    "security": "tls",
    "tlsSettings": {
      "certificates": [{
        "certificateFile": "/etc/letsencrypt/live/myserver.com/fullchain.pem",
        "keyFile": "/etc/letsencrypt/live/myserver.com/privkey.pem"
      }]
    },
    "wsSettings": {
      "path": "/ray",
      "headers": {}
    }
  }
}],
"outbounds": [{
  "protocol": "freedom"
}]
}
```

📞 If You Have Problems:

Information I Need:

1. What error do you see?
2. On which device (mobile/computer)?
3. Which method are you using (Reality or WebSocket)?
4. What is server log error? (command `journalctl -u xray -n 30`)

Immediate Help:

- Reality method is simpler and better
 - Make sure to copy UUID and keys correctly
 - Use port 443
 - Enable Allow Insecure in client
-

✓ Summary of Main Steps:

1. **Buy server** (DigitalOcean/Hetzner)
2. **Install Xray** (install command)
3. **Create config file** (with nano)
4. **Generate UUID and keys** (two commands: cat and x25519)
5. **Start service** (systemctl)
6. **Install on mobile** (v2rayNG)
7. **Test** (ipleak.net)